

DEVELOPMENT OF THE FINITE DIFFERENCE METHOD FOR VORTICITY FLOW SIMULATION USING THE FOURTH-ORDER RUNGE-KUTTA METHOD

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Abstract

The finite difference method is a numerical method to solve partial differential equations. The Forward Time Centered Space (FTCS) is an explicit finite difference method employed to solve parabolic partial differential equations. However, the FTCS scheme has a strict stability condition, requiring a very small time step. To address this limitation, this study integrates the centered difference scheme with fourth-order Runge-Kutta method (RK4), resulting in the RK4-CS scheme. The RK4-CS scheme is applied to simulate vorticity flow in a lid-driven cavity system. Validation is conducted by comparing the flow structures and computing error values using the Mean Absolute Percentage Error (MAPE) with the results from prior studies. The simulation results show that the vorticity flow structures obtained from the RK4-CS scheme are consistent with the results of prior studies. The MAPE values obtained from the RK4-CS scheme with time steps exceeding the stability limit of the FTCS scheme under the laminar regime is 2.3013%.

Keywords: Finite Difference Method, Fourth-Order Runge-Kutta Method, Lid-Driven Cavity, Stability Analysis, Vorticity Flow.

1 Introduction

Fluid flow is the continuous movement of a fluid substance in response of pressure variations or external forces [1]. This phenomenon is fundamentally governed by the conservation laws of mass and Newton's second law. Newton's second law applied to a fluid element yields a nonlinear partial differential equation known as the Navier-Stokes equations [2]. The solutions of the Navier-Stokes equations describe the velocity vector as a function of position and time. Due to nonlinear properties, the Navier-Stokes cannot be solve analytically. Hence, numerical methods are employed to obtain approximate solutions.

Finite Difference Method (FDM) is a numerical methods for solving partial differential equations. The problem domain is divided into discrete points called grid. Each derivative term in the differential equation is approximated using a Taylor series expansion. The partial differential equation is transformed into a system of linear equations known as the finite difference equations. The Forward Time Centered Space (FTCS) scheme is an explicit finite difference method for solving parabolic partial differential equations [3].

Several prior studies have investigated vorticity flows utilizing alternative method and various approaches [3]. *Ghia et al.* applied a multigrid approach to simulate high Reynolds number flows [7]. Meanwhile, *Erturk et al.* employed high grid resolutions to simulate flow conditions under identical Reynolds numbers [8]. Despite its relative simplicity in implementation, the FTCS scheme has strict stability condition, requiring very small time steps. This constraint has direct implications for computational cost increases significantly.

To address this limitation, this study investigates the development of a numerical scheme that integrates the Centered Space (CS) scheme as the spatial discretization with the fourth-order Runge-Kutta (RK4) method as the temporal integration, yielding the RK4-CS scheme. The RK4-CS scheme is applied to simulate vorticity flow in a lid-driven cavity system. Validation is conducted by comparing the flow structures and computing error values using the Mean Absolute Percentage Error (MAPE) with the results from prior studies. Simulation is conducted using Python programming language.

2 Problem Formulation

2.1 Governing Equations

The Navier-Stokes equations are derived from Newton's second law of motion applied to fluid flow. These equations are obtained from the Cauchy momentum equation for incompressible flow [4, 6]:

$$\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{v} + \mathbf{g} \quad (1)$$

where ρ is the fluid density, $\mathbf{v}$ is the velocity vector, p is the pressure, $\mathbf{g}$ is the gravitational acceleration, ν is the kinematic viscosity.

In the two-dimensional flow, the stream function (ψ) is a scalar function represents streamlines in incompressible flow. The relation between stream function and velocity components is:

$$\begin{aligned} u &= \frac{\partial \psi}{\partial y}, \\ v &= -\frac{\partial \psi}{\partial x} \end{aligned} \quad (2)$$

Vorticity (ω) is represents the local rotation of a fluid element. Vorticity is defined as the curl of the velocity vector:

$$\omega = \nabla \times \mathbf{v} \quad (3)$$

In a three-dimensional Cartesian coordinate system, the vorticity is expressed as:

$$\boldsymbol{\omega} = \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \mathbf{i} - \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \mathbf{j} + \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \mathbf{k} \quad (4)$$

In the two-dimensional flow, where the velocity component of z -direction (w) is zero and all velocity components are independent of z , Equation (4) is reduced to the scalar vorticity component in the z -direction [4, 6]:

$$\omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \quad (5)$$

Substituting the stream function into the two-dimensional vorticity definition yields the Poisson equation for the stream function [6, 7]:

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = -\omega \quad (6)$$

To derive the vorticity transport equation, the curl operator is applied to both sides of the Navier-Stokes equations. By eliminating the pressure and gravitational terms, the vorticity transport equation in two-dimensional is [6, 7]:

$$\frac{\partial \boldsymbol{\omega}}{\partial t} + \mathbf{v} \cdot \nabla \boldsymbol{\omega} = (\boldsymbol{\omega} \cdot \nabla) \mathbf{v} + \nu \nabla^2 \boldsymbol{\omega} \quad (7)$$

In the two-dimensional flow, the vortex stretching term $(\boldsymbol{\omega} \cdot \nabla) \mathbf{v}$ vanishes. By expressing the vorticity transport equation in terms of stream function, it simplified to [6]:

$$\frac{\partial \omega}{\partial t} + \frac{\partial \psi}{\partial y} \frac{\partial \omega}{\partial x} - \frac{\partial \psi}{\partial x} \frac{\partial \omega}{\partial y} = \nu \left(\frac{\partial^2 \omega}{\partial x^2} + \frac{\partial^2 \omega}{\partial y^2} \right) \quad (8)$$

Combining the Poisson equation and the vorticity transport equation yields the stream function-vorticity equation:

$$\begin{cases} \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = -\omega, \\ \frac{\partial \omega}{\partial t} + \frac{\partial \psi}{\partial y} \frac{\partial \omega}{\partial x} - \frac{\partial \psi}{\partial x} \frac{\partial \omega}{\partial y} = \nu \left(\frac{\partial^2 \omega}{\partial x^2} + \frac{\partial^2 \omega}{\partial y^2} \right) \end{cases} \quad (9)$$

Equation (9) as the mathematical model for simulating vorticity flow [6, 7, 8].

2.2 Initial and Boundary Conditions

The lid-driven cavity is a fluid flow model inside an enclosed cavity which the top wall moves at a constant velocity, while the remaining walls are held stationary. The configuration of this system is shown in Figure 1.

At the initial condition, the fluid is assumed rest throughout the domain Ω . The initial conditions of the system are [6]:

$$u_\Omega = v_\Omega = \psi_\Omega = \omega_\Omega = 0, (x, y) \in \Omega, t = 0. \quad (10)$$

Along all domain boundaries ($\partial\Omega$), the fluid is assumed respect to the wall velocity (no-slip condition). Accordingly, the stream function is zero on all walls. The vorticity at the walls is:

$$\omega_{\partial\Omega} = - \left. \frac{\partial^2 \psi}{\partial n^2} \right|_{\partial\Omega} - \left. \frac{\partial^2 \psi}{\partial s^2} \right|_{\partial\Omega} \quad (11)$$

where n and s are the wall-normal and wall-tangential directions, respectively. Illustration of grid in the around of walls is shown in Figure 2. To evaluate the second derivative of the stream

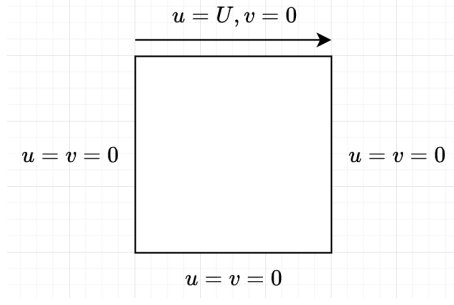


Figure 1: Configuration of the lid-driven cavity system.

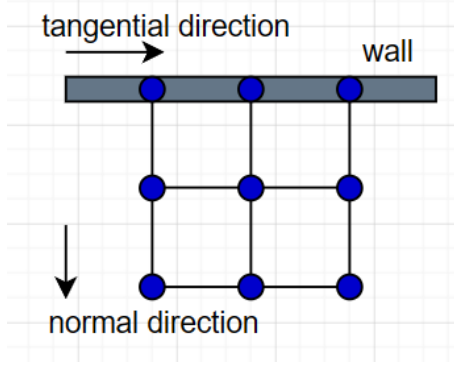


Figure 2: Grid points around the wall.

function with respect to n , a Taylor series expansion is applied in the wall-normal direction:

$$\left. \frac{\partial^2 \psi}{\partial n^2} \right|_{\partial \Omega} \approx \frac{2}{\Delta n^2} (\psi_{\partial \Omega + \Delta n} - \psi_{\partial \Omega}) - \frac{2}{\Delta n} \left. \frac{\partial \psi}{\partial n} \right|_{\partial \Omega} \quad (12)$$

By substituting the first-order wall-normal derivative of the stream function with the wall-tangential velocity (U_{wall}), Equation (12) can be rewritten as:

$$\left. \frac{\partial^2 \psi}{\partial n^2} \right|_{\partial \Omega} \approx \frac{2}{\Delta n^2} \psi_{\partial \Omega + \Delta n} \mp \frac{2}{\Delta n} U_{\partial \Omega} \quad (13)$$

Similarly, to evaluate the second derivative of the stream function with respect to s , a Taylor series expansion is applied in the wall-tangential direction:

$$\left. \frac{\partial^2 \psi}{\partial s^2} \right|_{\partial \Omega} \approx \frac{2}{\Delta s^2} (\psi_{\partial \Omega + \Delta s} - \psi_{\partial \Omega}) - \frac{2}{\Delta s} \left. \frac{\partial \psi}{\partial s} \right|_{\partial \Omega} \quad (14)$$

Since the stream function remains constant along the domain boundary and the wall-normal velocity is zero, we obtain:

$$\left. \frac{\partial^2 \psi}{\partial s^2} \right|_{\partial \Omega} = 0 \quad (15)$$

By substituting the second derivative of the stream function into Equation (11) yields the vorticity at the walls:

$$\omega_{\partial \Omega} = -\frac{2}{\Delta n^2} \psi_{\partial \Omega + \Delta n} \pm \frac{2}{\Delta n} U_{\partial \Omega} \quad (16)$$

The positive sign (+) applies when the wall-normal direction is aligned with the positive coordinate axes, while the negative sign (−) is applied in the opposite case. Equation (16) is applied to determine the vorticity boundary conditions on all walls. The boundary conditions of the system are [6, 8]:

Table 1: The boundary condition of *lid-driven cavity* system.

Dinding	Velocity component	Stream function	Vorticity
Left wall ($\partial\Omega_1$)	$u = 0, v = 0$	$\psi = 0$	$\omega = -\frac{2\psi_{\partial\Omega_1+\Delta x}}{\Delta x^2}$
Right wall ($\partial\Omega_2$)	$u = 0, v = 0$	$\psi = 0$	$\omega = -\frac{2\psi_{\partial\Omega_2-\Delta x}}{\Delta x^2}$
Bottom wall ($\partial\Omega_3$)	$u = 0, v = 0$	$\psi = 0$	$\omega = -\frac{2\psi_{\partial\Omega_3+\Delta y}}{\Delta y^2}$
Top wall ($\partial\Omega_4$)	$u = U, v = 0$	$\psi = 0$	$\omega = -\frac{2\psi_{\partial\Omega_4-\Delta y}}{\Delta y^2} - \frac{2U}{\Delta y}$

3 Discretization Method

3.1 FTCS Scheme

The FTCS scheme is based on a forward difference approximation for the time derivative and a centered difference approximation for the spatial derivatives. Both approximations are derived through Taylor series expansion [3, 6]. The first-order spatial derivative approximation is:

$$\begin{aligned}\left.\frac{\partial u}{\partial x}\right|_{i,j} &\approx \frac{u_{i+1,j} - u_{i-1,j}}{2\Delta x}, \\ \left.\frac{\partial u}{\partial y}\right|_{i,j} &\approx \frac{u_{i,j+1} - u_{i,j-1}}{2\Delta y}\end{aligned}\tag{17}$$

The second-order spatial derivative approximation is:

$$\begin{aligned}\left.\frac{\partial^2 u}{\partial x^2}\right|_{i,j} &\approx \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{\Delta x^2} \\ \left.\frac{\partial^2 u}{\partial y^2}\right|_{i,j} &\approx \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{\Delta y^2}\end{aligned}\tag{18}$$

The first-order time derivative approximation is:

$$\left.\frac{\partial u}{\partial t}\right|_{i,j}^n \approx \frac{u_{i,j}^{n+1} - u_{i,j}^n}{\Delta t}\tag{19}$$

Applying the centered difference approximation to the Poisson equation yields:

$$\begin{aligned}\psi_{i,j}^{(k+1)} = \frac{1}{2(\Delta x^2 + \Delta y^2)} &\left(\Delta x^2 \Delta y^2 \omega_{i,j} + \Delta x^2 (\psi_{i,j+1}^{(k)} + \psi_{i,j-1}^{(k)}) \right. \\ &\left. + \Delta y^2 (\psi_{i+1,j}^{(k)} + \psi_{i-1,j}^{(k)}) \right)\end{aligned}\tag{20}$$

To accelerate the its convergence rate, the Point Successive Over-Relaxation (PSOR) method is applied to Equation (20) [12]:

$$\begin{aligned}\psi_{i,j}^{(k+1)} = (1 - \beta)\psi_{i,j}^{(k)} &+ \frac{\beta}{2(\Delta x^2 + \Delta y^2)} \left(\Delta x^2 \Delta y^2 \omega_{i,j} \right. \\ &\left. + \Delta x^2 (\psi_{i,j+1}^{(k)} + \psi_{i,j-1}^{(k)}) + \Delta y^2 (\psi_{i+1,j}^{(k)} + \psi_{i-1,j}^{(k)}) \right)\end{aligned}\tag{21}$$

where β is the relaxation parameter. The vorticity transport equation is discretized using the FTCS scheme yields:

$$\begin{aligned}\omega_{i,j}^{n+1} = & \nu \Delta t \left(\frac{\omega_{i+1,j}^n - 2\omega_{i,j}^n + \omega_{i-1,j}^n}{\Delta x^2} + \frac{\omega_{i,j+1}^n - 2\omega_{i,j}^n + \omega_{i,j-1}^n}{\Delta y^2} \right) \\ & - \Delta t \left(\frac{\psi_{i,j+1}^n - \psi_{i,j-1}^n}{2\Delta y} \right) \left(\frac{\omega_{i+1,j}^n - \omega_{i-1,j}^n}{2\Delta x} \right) \\ & + \Delta t \left(\frac{\psi_{i+1,j}^n - \psi_{i-1,j}^n}{2\Delta x} \right) \left(\frac{\omega_{i,j+1}^n - \omega_{i,j-1}^n}{2\Delta y} \right) + \omega_{i,j}^n\end{aligned}\quad (22)$$

3.2 RK4-CS Scheme

The RK4-CS scheme is a hybrid method that integrates the CS scheme for spatial discretization with the RK4 method for temporal integration. The RK4 method is a numerical scheme for solving first-order ordinary differential equations (ODE) [9, 10]. The general form of a first-order ODE is:

$$\frac{d\omega}{dt} = f(t, \omega) \quad (23)$$

The numerical solution of a first-order ODE using RK4 method is [9, 10]:

$$\omega^{n+1} = \omega^n + \frac{1}{6} \Delta t (k_1 + 2k_2 + 2k_3 + k_4) \quad (24)$$

where

$$\begin{aligned}k_1 &= f(t, \omega) \\ k_2 &= f\left(t + \frac{\Delta t}{2}, \omega + \frac{k_1}{2} \Delta t\right) \\ k_3 &= f\left(t + \frac{\Delta t}{2}, \omega + \frac{k_2}{2} \Delta t\right) \\ k_4 &= f(t + \Delta t, \omega + k_3 \Delta t)\end{aligned}$$

The vorticity transport equation is spatially discretized using the centered difference approximation:

$$\begin{aligned}\frac{d\omega_{i,j}}{dt} = & \nu \left(\frac{\omega_{i+1,j} - 2\omega_{i,j} + \omega_{i-1,j}}{\Delta x^2} + \frac{\omega_{i,j+1} - 2\omega_{i,j} + \omega_{i,j-1}}{\Delta y^2} \right) \\ & - \left(\frac{\psi_{i,j+1} - \psi_{i,j-1}}{2\Delta y} \right) \left(\frac{\omega_{i+1,j} - \omega_{i-1,j}}{2\Delta x} \right) \\ & + \left(\frac{\psi_{i+1,j} - \psi_{i-1,j}}{2\Delta x} \right) \left(\frac{\omega_{i,j+1} - \omega_{i,j-1}}{2\Delta y} \right)\end{aligned}\quad (25)$$

Equation (25) is solved temporally using the RK4 method.

3.3 Conversion of the Stream Function to Velocity Components

The stream function is converted into velocity components. By applying the centered difference approximation to the stream function definition, the relationship between these variables is:

$$\begin{aligned}u_{i,j}^n &\approx \frac{\psi_{i,j+1}^n - \psi_{i,j-1}^n}{2\Delta y} \\ v_{i,j}^n &\approx -\frac{\psi_{i+1,j}^n - \psi_{i-1,j}^n}{2\Delta x}\end{aligned}\quad (26)$$

4 Stability Analysis

A stability analysis is conducted to ensure that the numerical solution does not diverge as time and iterations progress. This analysis is performed using the Von Neumann method by representing the solution as a complex exponential form [3]:

$$\omega_{j,l}^n = \rho^n e^{i(jk_x \Delta x + lk_y \Delta y)} \quad (27)$$

where ρ is the amplification factor and k is the sinusoidal wave number. A numerical scheme is stable if the amplification factor satisfies the condition $|\rho| \leq 1$.

The numerical stability limits are expressed in two dimensionless parameters, the diffusion number (r) and the Courant number or Courant-Friedrich-Lewy/CFL number (C). These parameters are defined as:

$$r = \frac{\nu \Delta t}{\Delta x^2} \quad (28)$$

$$C = \frac{U \Delta t}{\Delta x} \quad (29)$$

4.1 FTCS Scheme

To perform the stability analysis of Equation (23), the stream function is substituted by constant velocity components (U and V) [11]. Equation (23) is transformed into the advection-diffusion form:

$$\begin{aligned} \omega_{i,j}^{n+1} = & \nu \Delta t \left(\frac{\omega_{i+1,j}^n - 2\omega_{i,j}^n + \omega_{i-1,j}^n}{\Delta x^2} + \frac{\omega_{i,j+1}^n - 2\omega_{i,j}^n + \omega_{i,j-1}^n}{\Delta y^2} \right) \\ & - U \Delta t \left(\frac{\omega_{i+1,j}^n - \omega_{i-1,j}^n}{2\Delta x} \right) + V \Delta t \left(\frac{\omega_{i,j+1}^n - \omega_{i,j-1}^n}{2\Delta y} \right) + \omega_{i,j}^n \end{aligned} \quad (30)$$

Substituting the complex exponential solution into Equation (30) yields the amplification factor of the FTCS scheme:

$$\begin{aligned} \rho = & 1 - 4 \left(\frac{\nu \Delta t}{\Delta x^2} \sin^2 \left(\frac{k_x \Delta x}{2} \right) + \frac{\nu \Delta t}{\Delta y^2} \sin^2 \left(\frac{k_y \Delta y}{2} \right) \right) \\ & - i \left(\frac{U \Delta t}{\Delta x} \sin(k_x \Delta x) + \frac{V \Delta t}{\Delta y} \sin(k_y \Delta y) \right) \end{aligned} \quad (31)$$

Assuming a uniform grid ($\Delta x = \Delta y$) and constant velocity components (U and V) as the top wall velocity in the lid-driven cavity system, Equation (31) is simplified to:

$$\rho = 1 - 8r \sin^2 \left(\frac{\theta}{2} \right) - iC \sin(\theta) \quad (32)$$

where $\theta = k\Delta x$. Based on the Von Neumann stability criterion, the FTCS scheme is stable if:

$$-4r + 16r^2 \sin^2 \left(\frac{\theta}{2} \right) + C^2 \cos^2 \left(\frac{\theta}{2} \right) \leq 0 \quad (33)$$

By examining the extreme values of the trigonometric functions in Equation (33), when the sine function reaches its maximum ($\theta \rightarrow \pi/2$), the stability of the diffusion condition is:

$$r \leq \frac{1}{4} \quad (34)$$

When the cosine function reaches its maximum ($\theta \rightarrow 0$), the stability of the advection-diffusion condition is:

$$C^2 \leq 4r \quad (35)$$

The stability region of the FTCS scheme is the intersection of these two conditions, shown in Figure 3.

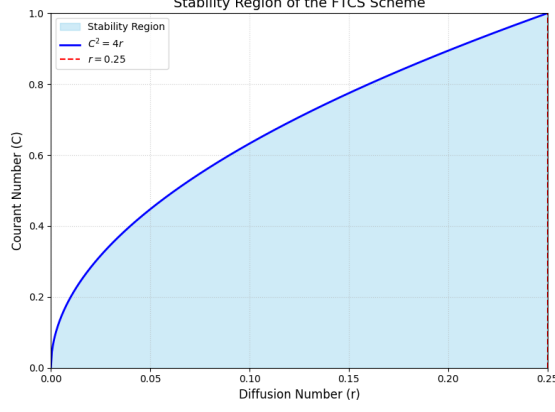


Figure 3: Stability region of the FTCS scheme.

4.2 RK4-CS Scheme

To perform the stability analysis of the RK4-CS scheme, the linear characteristic differential equation is defined as:

$$\frac{d\omega}{dt} = \lambda\omega \quad (36)$$

The numerical solution of Equation (36) obtained using the RK4 method is:

$$\omega^{n+1} = \omega^n + (\lambda\Delta t)\omega^n + \frac{(\lambda\Delta t)^2\omega^n}{2!} + \frac{(\lambda\Delta t)^3\omega^n}{3!} + \frac{(\lambda\Delta t)^4\omega^n}{4!} \quad (37)$$

Substituting the complex exponential solution into Equation (37) yields the amplification factor of the RK4-CS scheme [10]:

$$\rho = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} \quad (38)$$

where $z = \lambda\Delta t$ is a complex parameter. According to the Von Neumann stability criterion, the RK4-CS scheme is stable if the condition $|\rho(z)| \leq 1$ is satisfied. Substituting the complex exponential solution into Equation (25) yields:

$$\begin{aligned} \frac{d\omega}{dt} = & -\rho^n e^{i(jk_x\Delta x + lk_y\Delta y)} \left(\frac{4\nu}{\Delta x^2} \sin^2\left(\frac{k_x\Delta x}{2}\right) + \frac{4\nu}{\Delta y^2} \sin^2\left(\frac{k_y\Delta y}{2}\right) \right. \\ & \left. + i\frac{U}{\Delta x} \sin(k_x\Delta x) + i\frac{V}{\Delta y} \sin(k_y\Delta y) \right) \end{aligned} \quad (39)$$

Comparing Equation (39) with the characteristic differential equation yields:

$$\begin{aligned} z = & -\frac{4\nu\Delta t}{\Delta x^2} \sin^2\left(\frac{k_x\Delta x}{2}\right) - \frac{4\nu\Delta t}{\Delta y^2} \sin^2\left(\frac{k_y\Delta y}{2}\right) \\ & - i\left(\frac{U\Delta t}{\Delta x} \sin(k_x\Delta x) + \frac{V\Delta t}{\Delta y} \sin(k_y\Delta y)\right) \end{aligned} \quad (40)$$

Assuming a uniform grid and constant velocity parameters, the complex parameter is simplified to:

$$z = -8r \sin^2\left(\frac{k\Delta x}{2}\right) - iC \sin(k\Delta x) \quad (41)$$

By examining the extreme values of the trigonometric functions in Equation (41), when the sine function reaches its maximum ($\theta \rightarrow \pi/2$) yields:

$$z = -4r - iC \quad (42)$$

The stability region of the RK4-CS scheme is shown in Figure 4.

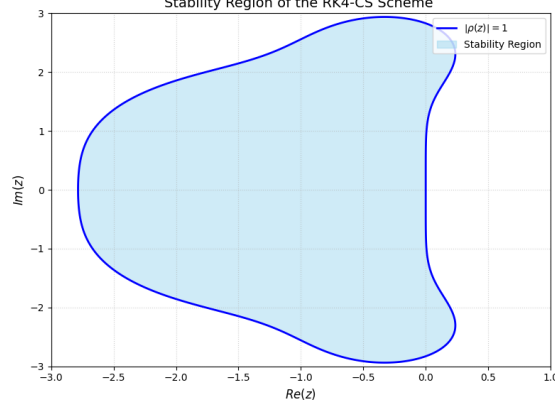


Figure 4: Stability region of the RK4-CS scheme.

5 Algorithm

The procedure for solving the stream function-vorticity equation is carried out through the following algorithmic steps:

1. Apply the initial conditions for the stream function and vorticity.
2. Solve Equation (21) to obtain the values of stream function until the convergence criterion $(|\psi^{(k+1)} - \psi^{(k)}| \leq \epsilon)$ is satisfied.
3. Compute the vorticity values by solving Equation (22) for the FTCS scheme or Equation (25) for the RK4-CS scheme.
4. Apply the boundary conditions for both the stream function and vorticity.
5. Convert the stream function solution into velocity components using Equation (26).

6 Results and Discussions

Simulations are performed under the laminar regime by varying the time step. The fixed parameters including the Reynolds number ($Re = 1000$), the wall length ($L = 1\text{ m}$), the top wall velocity ($U = 1\text{ m s}^{-1}$), the fluid kinematic viscosity ($\nu = 10^{-3}\text{ m}^2\text{ s}^{-1}$), the grid resolution of 100×100 , the error tolerance ($\epsilon = 10^{-4}$), and the relaxation parameter ($\beta = 1.75$).

In the first scenario, the simulation is performed with time step ($\Delta t = 2.5 \times 10^{-3}$) which satisfies the stability criteria of both schemes. The vorticity flow structure obtained from FTCS scheme is shown in Figure (5), similar to RK4-CS scheme. In the second scenario, the simulation is performed with the time step increased to ($\Delta t = 10^{-2}$) which only satisfies the stability criterion of the RK4-CS scheme. The vorticity flow structure obtained from the RK4-CS scheme is shown in Figure (6).

Figure (5) and (6) shows three vortices located at the cavity center, the lower-left corner, and the lower-right corner of the domain. The primary vortex initially forms near the top wall and moves toward the center of the cavity until a steady-state condition is reached. Simultaneously, secondary and tertiary vortices are develop at the lower-right and lower-left corners, respectively. The vorticity flow structures from these two scenarios are in consistent with the results reported by *Erturk et al.* [8]

Validation is carried out by comparing the flow structure and calculating the error value using the *Mean Absolute Percentage Error* (MAPE) with the results reported by *Erturk et al.* [8]. The MAPE calculation is based on the u -velocity along the vertical line and the v -velocity

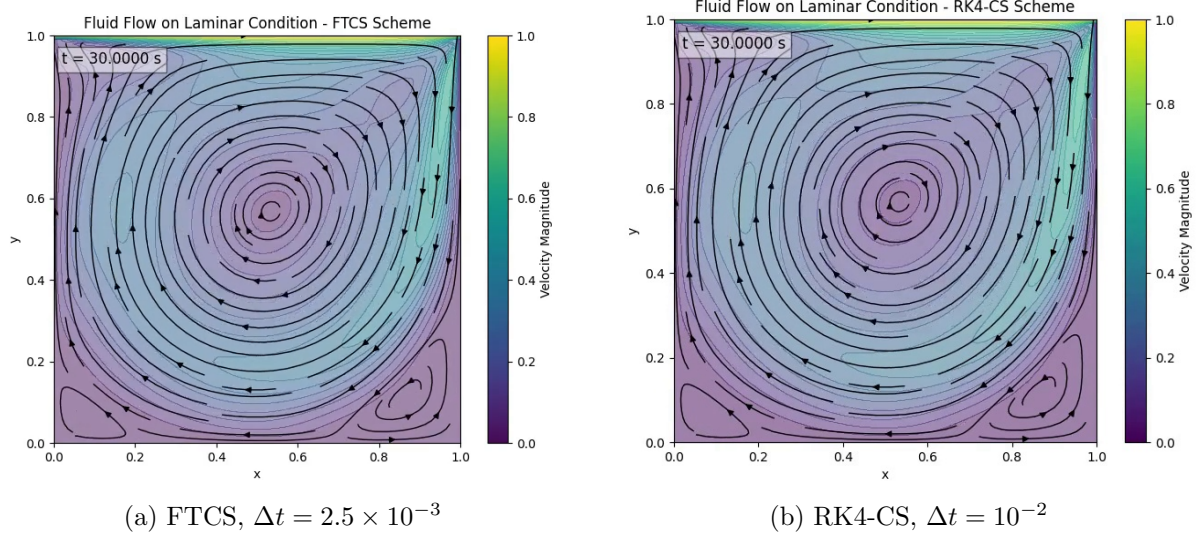


Figure 5: Vorticity flow structures in the laminar regime at $t = 30$ s.

along the horizontal line, which through the center of the domain. A comparison of the velocity component profiles in the laminar regime is shown in Figure 6.

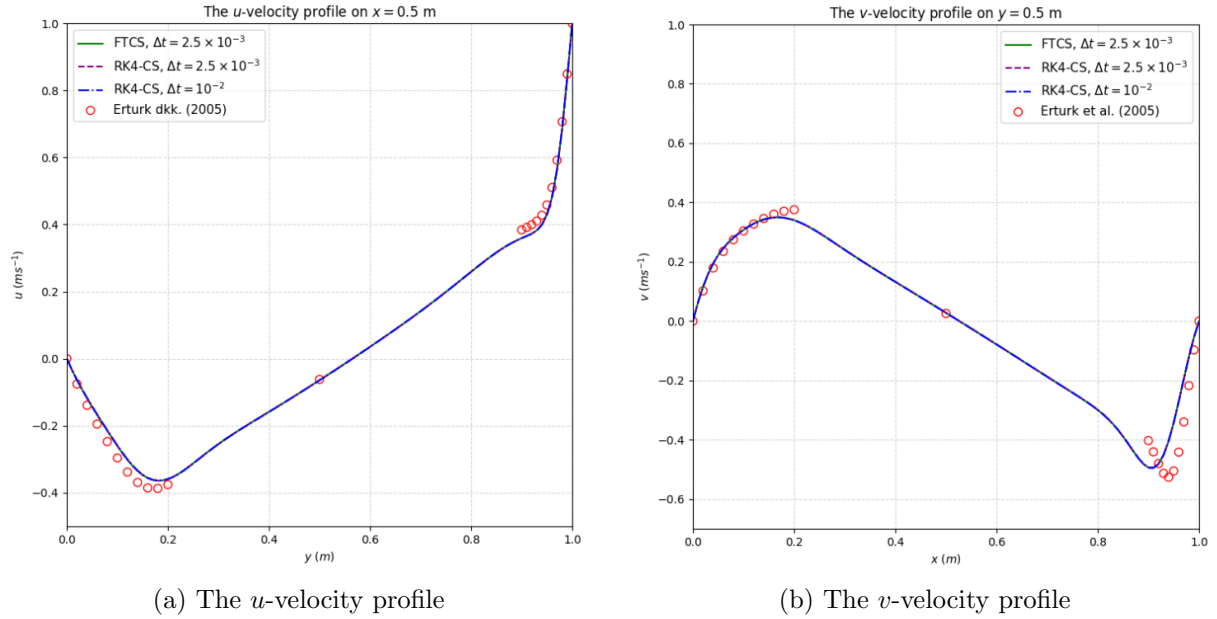


Figure 6: The comparison of the velocity component profile at $t = 30$ s.

The MAPE values of FTCS, RK4-CS ($\Delta t = 2.5 \times 10^{-3}$), and RK4-CS ($\Delta t = 10^{-2}$) schemes are 2.2855%, 2.2850%, and 2.3013%, respectively.

7 Conclusions

The structures of the vorticity flow obtained from the RK4-CS scheme in the laminar regime is consistent with the FTCS scheme and the results reported by Erturk et al. [8]. The numerical solution obtained from the RK4-CS scheme is stable at time steps beyond the stability criteria of the FTCS scheme, thereby expanding the stability limits of the FTCS scheme. In the laminar regime, the MAPE values of RK4-CS scheme at time steps exceeding the FTCS stability criteria

is 4.5303%.

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